| Program : **Diploma in Polymer Technology** | |
|---|---|
| Course Code : **6099** | Course Title: **Fibers and Composite Workshop** |
| Semester : **6** | Credits: **1.5** |
| Course Category: **Program Elective** | |
| Periods per week: **3 (L:0, T:0, P:3)** | Periods per semester: **45** |

**Course Objectives:**

- To provide the skill for testing properties of fibre used for the manufacture of fibre reinforced plastics.
- To determine the curing characteristics of the resin used.
- To get hands on experience of the production of different FRP products.

**Course Prerequisites:**

| Topic | Course code | Course name | Semester |
|---|---|---|---|
| Basic Knowledge in chemistry | | Applied Chemistry | 1 |
| Basic Knowledge in polymer chemistry | | Polymer science | 3 |
| Basic knowledge in plastics and resins | | Plastic and composites | 5 |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive level |
|---|---|---|---|
| CO1 | Determine denier, tex, and linear density of fibres. | 10 | Applying |
| CO2 | Determine gel time, set time and cure time of the resin. | 9 | Applying |
| CO3 | Produce composite products by hand layup process and cast articles. | 24 | Applying |
| | Lab Exam | 2 | |

**CO-PO Mapping:**

| Course Outcome | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 3 | 3 | | 3 | | | |
| **CO2** | 3 | 3 | | | | | |
| **CO3** | 3 | | | 3 | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Description | Duration (Hours) | Cognitive level |
|---|---|---|---|
| **CO1** | **Determine denier, tex, and linear density of fibres.** | | |
| M1.01 | Determine Denier of Cotton, Wool, Nylon, Polyester | 5 | Applying |
| M1.02 | Determine Tex of Cotton, Wool, Nylon, Polyester. | 5 | Applying |
| **CO2** | **To determine gel time, set time and cure time of the resin.** | | |
| M2.01 | Determine the Gel time, set time and cure time of Polyester resin. | 3 | Applying |
| M2.02 | Determine the Gel time, set time and cure time of Polyester resin by varying dosage of catalyst. | 3 | Applying |
| M2.03 | Determine the Gel time, set time and cure time of Polyester resin by varying dosage of accelerator. | 3 | Applying |
| | Lab exam I | 1 | |
| **CO3** | **Produce composite products by hand layup process and cast articles.** | | |
| M3.01 | Perform the Hand layup process for different products (single layer, double layer with inner and outer finish ) | 10 | Applying |
| M3.02 | Perform the Hand layup process for different products with double side finish. | 10 | Applying |
| M3.03 | Carry out Casting process with Polyester resin | 4 | Applying |
| | Lab exam II | 1 | |

**Text / Reference:**

| T/R | Book Title/Author |
|---|---|
| T1 | FRP technology - R.G Weatherhead (Applied Science Publishers 1980 |
| R2 | Plastic Product Design Engineering Hand book – Sydney Levy, J Harry DuBois (Springer US,1984) |
| R3 | Plastic materials J.A Brydson( Butterworth Heinmann, 7th Edition |
| R4 | Plastics Engineering Handbook Of The Society Of The Plastics Industry – M. Berins, (Publisher - |
| R5 | Plastic Technology Hand book - William J Patton (Publisher - Reston |

**Online Resources:**

| Sl. No | Website Link |
|---|---|
| 1 | https://docs.woocommerce.com |
| 2 | Compositeslab.com. |
| 3 | www.sciencedirect.com |
| 4 | www.eppcomposites.com |